

Annex O (normative)

Requirements justification file (RJF) - DRD

O.1 DRD identification

O.1.1 Requirement identification and source document

This DRD is called from ECSS-E-ST-10, requirement 5.2.1e.

O.1.2 Purpose and objective

The requirement justification file (RJF) is a generic title referring to all documentation which:

- Records and describes the needs and the associated constraints resulting from the different trade-offs.
- Demonstrates how the requirements of the technical requirements specification (TS) (as defined in ECSS-E-ST-10-06 Annex A) at each level can satisfy the needs and the constraints of the TS of the level above.

NOTE A top level RJF document is established at the upper level of the project structure that is, according to ECSS-M-ST-10, the first-level customer situated at level 0 of the customer supplier network. For other levels, the RJF is part of the design justification file (DJF) (as defined in, Annex K).

O.2 Expected response

O.2.1 Scope and content

<1> Introduction

- a. The RJF shall contain a description of the purpose, objective, content and the reason prompting its preparation.

<2> Applicable and reference documents

- a. The RJF shall list the applicable and reference documents in support to the generation of the document.
- b. The RJF shall include the reference to the technical requirements specification.

<3> Selected concept/solution justification

- a. The RJF shall present the rationale for the selection of a concept for the technical requirements specification.

NOTE For justification, reference can be made to the system concept report of the considered project.

<4> Life profile justification for the selected concept/solution

- a. The RJF shall present and justify the life profile situations for the concept presented in the technical requirements specification.

NOTE For justification, reference can be made to the system concept report of the considered project where relevant.

<5> Environments and constraints justification

- a. The RJF shall present and justify the different environments and constraints for each life profile situations for the concept presented in the technical requirements specification.

NOTE For justification, reference can be made to the system concept report of the considered project where relevant.

<6> Technical requirements justification

- a. The RJF shall list all the technical requirements, and their identifier, expressed by the corresponding TS as they are organized in these documents.
- b. For each technical requirement the following information shall be provided:
 - 1. the justification of the requirement (i.e. justified source),
 - 2. the Entity or owner responsible for the requirement,
 - 3. if one technical requirement is induced by several sources, the reason of the specified performance,
 - 4. the justification of the quantified performance (such as the range, the approach used to determine the level, e.g. measure, estimation),
 - 5. the justification of the selected verification method.

<7> Requirement traceability

- a. The RJF shall present the requirement traceability between the technical requirements of the TS, and their justified source.

<8> Compliance matrix for COTS

- a. For COTS, the RJF shall contain a compliance matrix between the technical specification/characteristics of the COTS and the technical requirement expressed by the TS.

O.2.2 Special remarks

- a. The RJF may be part of the design justification file (DJF) (as defined in ECSS-E-ST-10 Annex K).